SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: INTERNET PROGRAMMING
COURSE CODE: CSA43

COURSE OUTCOME COVERED

CO-1: Develop well-structured, standards-compliant web pages using HTML/XHTML and XML, and apply Internet protocols and HTTP communication mechanisms for web content delivery. (L2)

Assessment Tool: Application-Based Questions

Weightage: 40%

QUESTIONS: **Total Marks: 30**

Code of Conduct:

I K. Lahari(192325118) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: K. Lahari

QUESTION 1: Online Symposium Portal

A college is developing an online symposium registration portal. The following HTML structure and HTTP interaction is observed:

```
<form action="/register" method="POST">
  <input type="text" name="name">
  <input type="email" name="email">
  <button>Submit</button>
</form>
```

When a student submits the form, the browser sends an HTTP request to the server, and the server responds accordingly.

Questions:

1. **Identify appropriate HTML5 semantic elements missing in the structure.**
2. **Modify the structure to make it standards-compliant.**
3. **Interpret the HTTP request generated during form submission.**
4. **Analyze the HTTP response cycle from server to browser.**
5. **Suggest improvements for usability and accessibility.**

ANSWER :

Code:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Online Symposium Registration</title>

  <style>
    body {
      font-family: Arial, sans-serif;
      background-color: #f0f4f8;
      margin: 0;
      padding: 20px;
    }

    header {
```

```
    background-color: #1769aa;
    color: white;
    text-align: center;
    padding: 20px;
}

main {
    max-width: 450px;
    margin: 30px auto;
    background-color: white;
    padding: 25px;
    border-radius: 10px;
}

label {
    display: block;
    margin-top: 15px;
    margin-bottom: 5px;
}

input {
    width: 100%;
    padding: 10px;
    box-sizing: border-box;
}

button {
    background-color: #1769aa;
    color: white;
    padding: 12px;
    width: 100%;
    border: none;
    margin-top: 20px;
    cursor: pointer;
    border-radius: 5px;
}

button:hover {
    background-color: #0d4777;
}

footer {
    text-align: center;
    margin-top: 20px;
```

```
        }

        #message {
            color: green;
            text-align: center;
            margin-top: 15px;
        }
    </style>
</head>

<body>

    <header>
        <h1>Online Symposium</h1>
        <p>College Symposium Registration Portal</p>
    </header>

    <main>
        <section>
            <h2>Student Registration</h2>

            <form id="registrationForm">
                <label for="name">Student Name:</label>
                <input type="text" id="name" name="name"
                       placeholder="Enter your name" required>

                <label for="email">Email Address:</label>
                <input type="email" id="email" name="email"
                       placeholder="Enter your email" required>

                <button type="submit">Register</button>
            </form>

            <p id="message" role="status"></p>
        </section>
    </main>

    <footer>
        <p>&copy; 2026 College Symposium</p>
    </footer>

    <script>
        const form = document.getElementById("registrationForm");
        const message = document.getElementById("message");
```

```
    form.addEventListener("submit", function(event) {
      event.preventDefault();

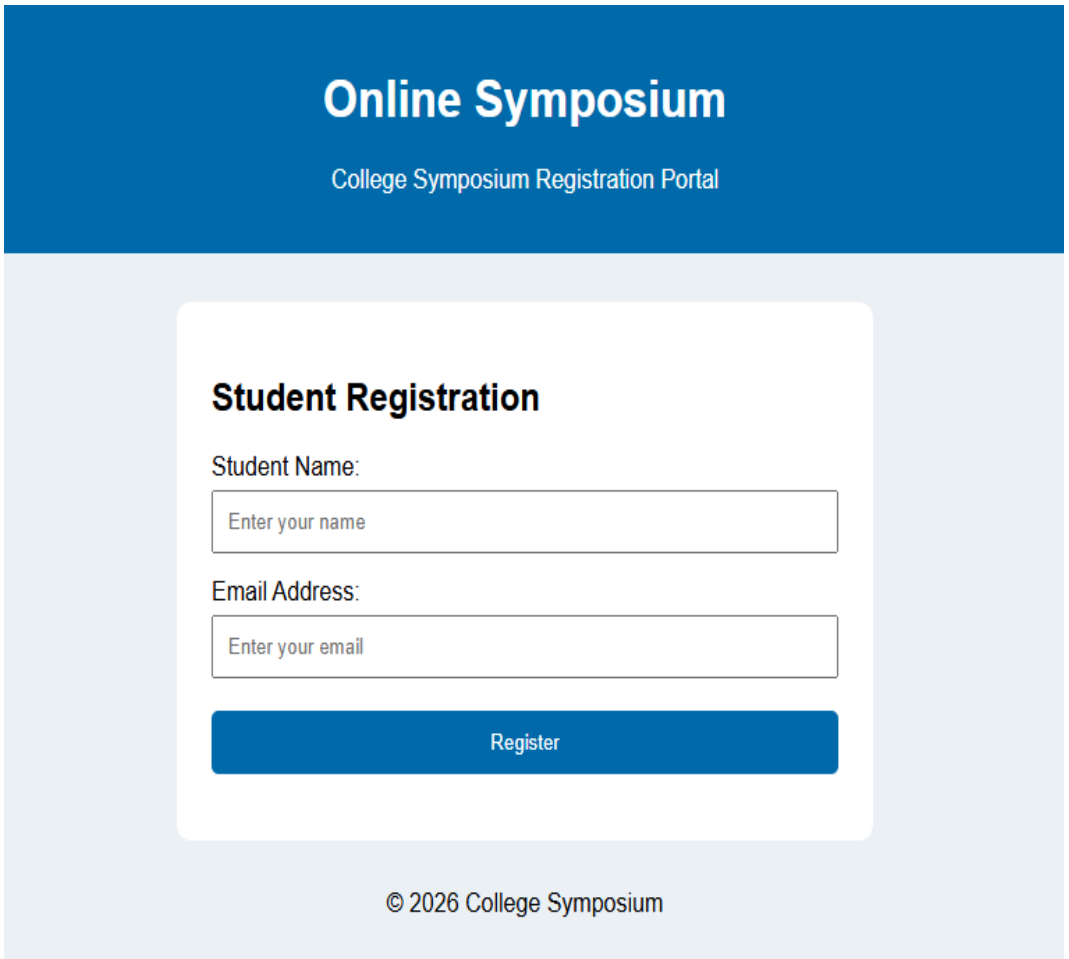
      const name = document.getElementById("name").value;

      message.textContent =
        "Registration successful! Welcome, " + name + ".";

      form.reset();
    });
  </script>

</body>
</html>
```

Output:



QUESTION 2: Cross-Browser Compatibility Issue

A company website shows inconsistent layout across browsers. The following HTML snippet is used:

```
<html>
<head>
<title>Company</title>
</head>
<body>
<div>
<h1>Welcome
<p>About us
</div>
</body>
</html>
```

Questions:

1. **Identify improper nesting and missing closing tags.**
2. **Analyze how different browsers may interpret this structure.**
3. **Identify missing semantic elements (e.g., header, section).**
4. **Provide a corrected W3C-compliant HTML structure.**
5. **Suggest techniques to ensure cross-browser compatibility.**

ANSWER :

Code:

```
<!DOCTYPE html>
<html lang="en">
<head>
   <meta charset="UTF-8">
   <meta name="viewport" content="width=device-width, initial-scale=1.0">
   <title>Company Website</title>

   <style>
      * {
         box-sizing: border-box;
      }
```

```
body {
    margin: 0;
    font-family: Arial, sans-serif;
    line-height: 1.6;
    color: #1e293b;
    background: #f8fafc;
}

header {
    background: #0f766e;
    color: white;
    text-align: center;
    padding: 35px 15px;
}

header h1 {
    margin: 0;
    font-size: 36px;
}

nav {
    background: #115e59;
    text-align: center;
    padding: 12px;
}

nav a {
    color: white;
    text-decoration: none;
    margin: 0 15px;
}

nav a:hover {
    text-decoration: underline;
}

main {
    max-width: 900px;
    margin: 30px auto;
```

```
        padding: 0 20px;
    }

    section {
        background: white;
        padding: 25px;
        margin-bottom: 20px;
        border-radius: 10px;
        box-shadow: 0 3px 12px #00000010;
    }

    h2 {
        color: #0f766e;
    }

    footer {
        background: #134e4a;
        color: white;
        text-align: center;
        padding: 18px;
    }

    @media (max-width: 600px) {
        header h1 {
            font-size: 28px;
        }

        nav a {
            display: inline-block;
            margin: 5px 8px;
        }
    }
  </style>
</head>

<body>

  <header>
    <h1>Welcome</h1>
    <p>Building Ideas into Solutions</p>
```

```
    </header>

    <nav aria-label="Main navigation">
      <a href="#home">Home</a>
      <a href="#about">About Us</a>
      <a href="#contact">Contact</a>
    </nav>

    <main id="home">
      <section id="about">
        <h2>About Us</h2>
        <p>
          Welcome to our company website.
          We focus on innovation, quality, and customer satisfaction.
        </p>
      </section>

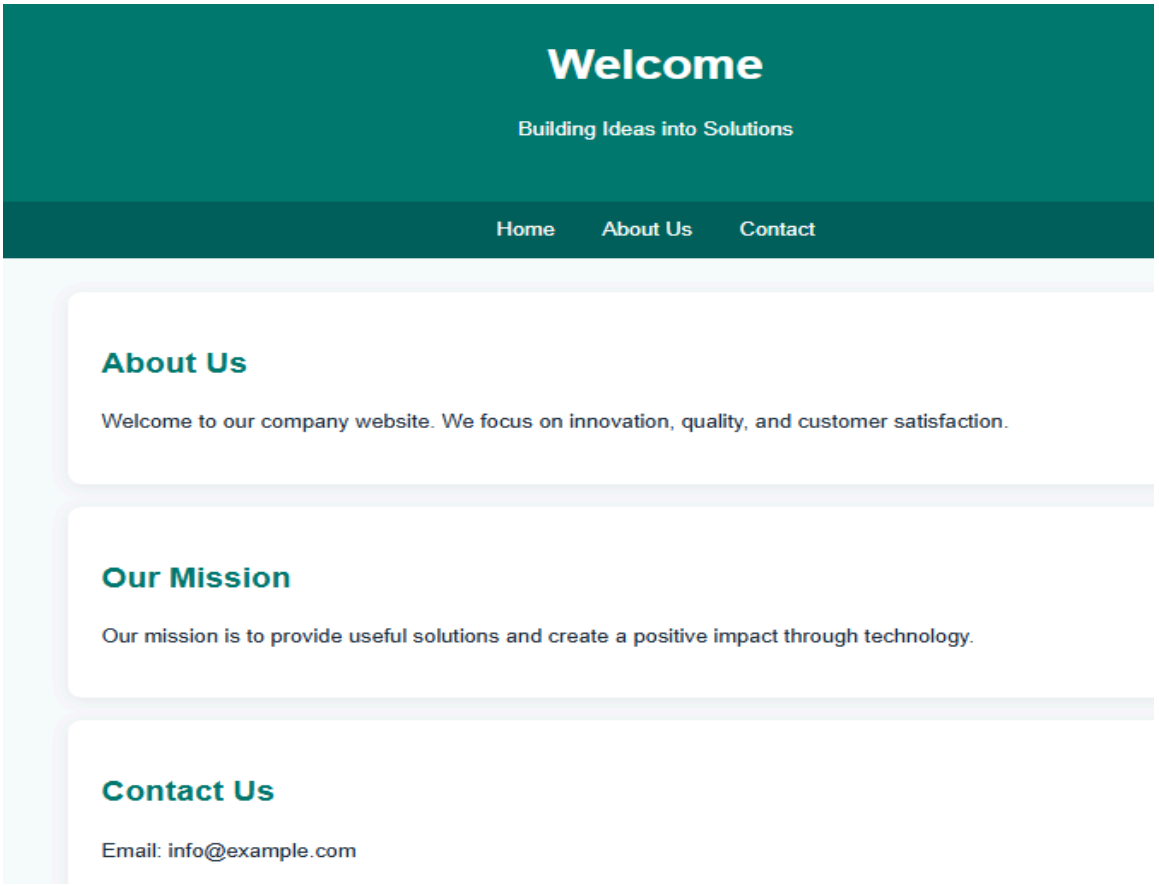
      <section>
        <h2>Our Mission</h2>
        <p>
          Our mission is to provide useful solutions
          and create a positive impact through technology.
        </p>
      </section>

      <section id="contact">
        <h2>Contact Us</h2>
        <p>Email: info@example.com</p>
      </section>
    </main>

    <footer>
      <p>&copy; 2026 Company Website. All rights reserved.</p>
    </footer>

</body>
</html>
```

Output :


Welcome
Building Ideas into Solutions
Home
About Us
Contact
About Us
Welcome to our company website. We focus on innovation, quality, and customer satisfaction.
Our Mission
Our mission is to provide useful solutions and create a positive impact through technology.
Contact Us
Email: info@example.com

QUESTION 3: Form Submission Failure

A web application fails to send form data to the server. The following configuration is observed:

```
<form action="/submit" method="GET">
  <input type="text" name="username">
  <input type="password" name="password">
  <button type="button">Login</button>
</form>
```

Questions:

1. **Identify issues in the form configuration.**
2. **Analyze why the HTTP request is not triggered.**
3. **Interpret the role of GET method in this scenario.**
4. **Propose a corrected implementation.**
5. **Suggest improvements for secure data transmission.**

ANSWER :

Code:

```
<!DOCTYPE html>
<html lang="en">
<head>
   <meta charset="UTF-8">
   <meta name="viewport" content="width=device-width, initial-scale=1.0">
   <title>Secure Login Form</title>

   <style>
      * {
         box-sizing: border-box;
      }

      body {
         margin: 0;
         min-height: 100vh;
         display: flex;
         justify-content: center;
```

```
    align-items: center;
    padding: 20px;
    font-family: Arial, sans-serif;
    background: linear-gradient(135deg, #dbeafe, #ede9fe);
    color: #1e293b;
}

main {
    width: 100%;
    max-width: 420px;
}

section {
    background: white;
    padding: 32px;
    border-radius: 14px;
    box-shadow: 0 8px 25px #00000015;
}

h1 {
    text-align: center;
    color: #4338ca;
    margin-top: 0;
}

.subtitle {
    text-align: center;
    color: #64748b;
    margin-bottom: 25px;
}

label {
    display: block;
    margin: 18px 0 7px;
    font-weight: bold;
}

input {
    width: 100%;
    padding: 12px;
```

```
            border: 1px solid #cbd5e1;
            border-radius: 6px;
            font-size: 16px;
        }

        input:focus {
            outline: 2px solid #a5b4fc;
            border-color: #4f46e5;
        }

        button {
            width: 100%;
            padding: 13px;
            margin-top: 24px;
            border: none;
            border-radius: 6px;
            background: #4338ca;
            color: white;
            font-size: 16px;
            cursor: pointer;
        }

        button:hover {
            background: #3730a3;
        }

        #message {
            text-align: center;
            margin-top: 18px;
            font-weight: bold;
        }

        footer {
            text-align: center;
            color: #64748b;
            margin-top: 18px;
            font-size: 13px;
        }
    </style>
</head>
```

```
<body>

  <main>
    <section>
      <h1>Welcome Back</h1>
      <p class="subtitle">Enter your details to continue.</p>

      <form id="loginForm"
          action="/submit" method="POST">

        <label for="username">Username</label>
        <input type="text" id="username"
            name="username"
            placeholder="Enter your username"
            autocomplete="username"
            required>

        <label for="password">Password</label>
        <input type="password" id="password"
            name="password"
            placeholder="Enter your password"
            autocomplete="current-password"
            required>

        <button type="submit">Login</button>
      </form>

      <p id="message" role="status" aria-live="polite"></p>
    </section>

    <footer>
      Secure Login Form Demonstration
    </footer>
  </main>

  <script>
    const form = document.getElementById("loginForm");
    const message = document.getElementById("message");
```

```
        form.addEventListener("submit", function(event) {
            event.preventDefault();

            message.style.color = "#b45309";
            message.textContent =
                "Demo only: connect a server to authenticate your login.";
        });
    </script>

</body>
</html>
```

